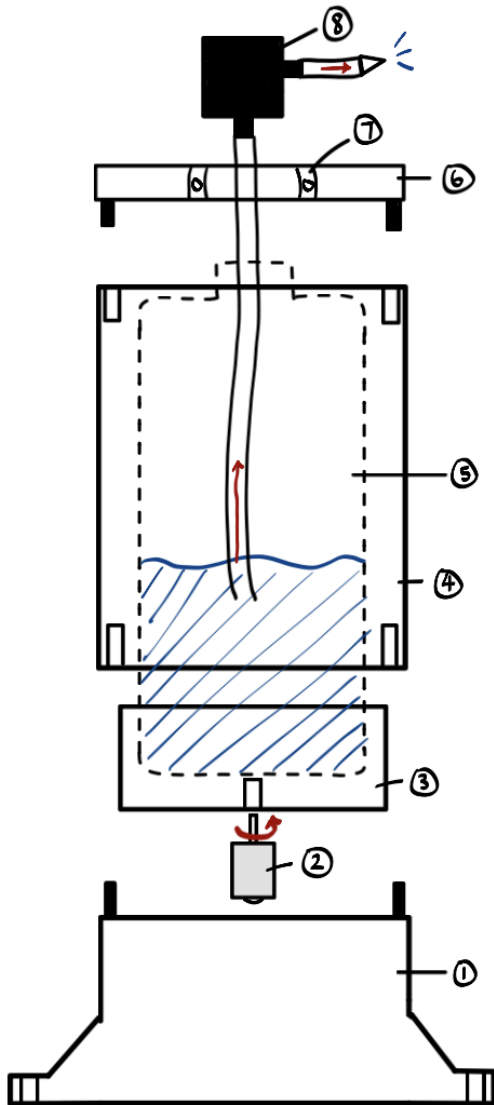


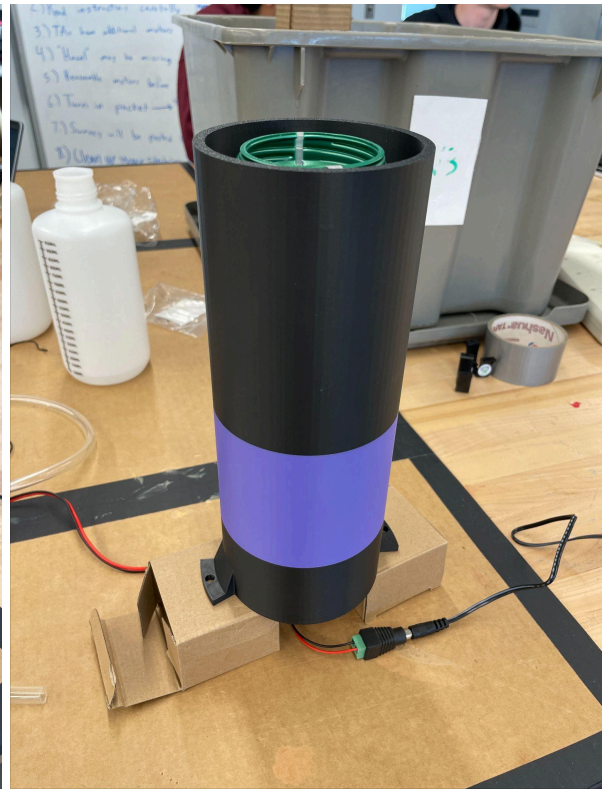
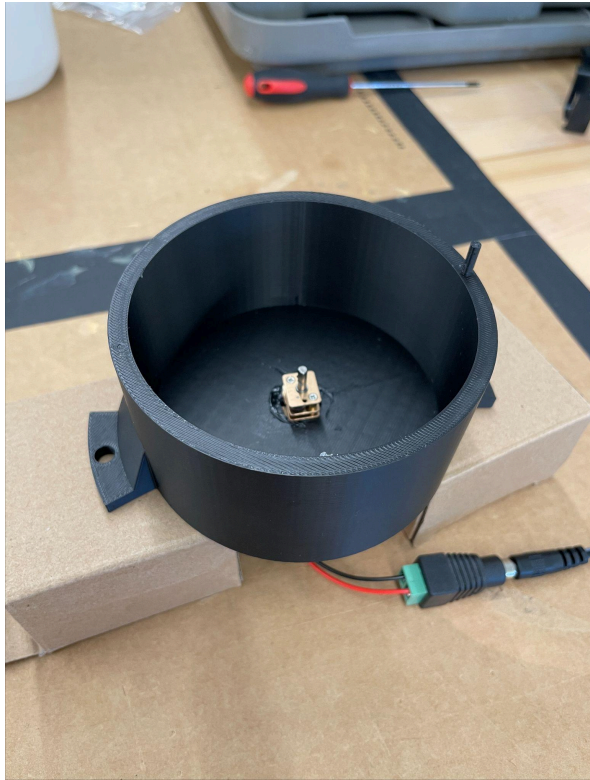
Targeted Oviposition Control for SLF: Functional Prototype

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Design Documentation:



Part	Fabrication Details/Notes
1. Base/Motor Compartment	3D printed; 12 mm hole hand-drilled through bottom to hold the DC motor
2. DC Motor	Micro N20 gear motor, 3-6V
3. Bottle Holder	3D printed
4. Outer Shell	3D printed
5. Bottle	32 oz HDPE plastic bottle; 2 mm was sawed off its height from the opening
6. Bearing Support	3D printed
7. Ball Bearing	McMaster-Carr (5972K217)
8. Water Pump	DC 12V submersible pump



Design Test:

Component Fit Test: Were we able to assemble our components to see where everything should fit in our next design?

This test was conducted to determine whether all components of the system could be properly assembled and positioned within the design. The goal was to verify that each part fits as intended and to identify any spatial or alignment issues before moving forward with the next design iteration. The results showed that the components could be assembled successfully, with one caveat: the bottle we purchased did not fit in our printed component. We were able to visualize working conditions by using another bottle, but we will make alterations in future iterations. Therefore, adjustments will be required to improve fit and stability.

Pump Function Test: Is the pump able to move water effectively out of our chamber into another bottle?

This test was designed to measure the flow rate of the pump as it moves the liquid from the chamber to the nozzle. We simulated these circumstances by filling the chamber with water and then pumping that water into another container (set volume) to see how long it would take (mL/sec). However, we soon concluded that our pump would not be effective in this endeavor. We discovered that the pump we had purchased was unable to self-prime, so it is not suitable for our system which is designed to turn on and off on its own.

Mixing Efficiency: Does our spinning, motor-driven mixing mechanism effectively mix the liquid inside the chamber?

This test was designed to determine the efficiency of our mechanism to actually mix the liquid inside the chamber. A drop of food coloring was added to the liquid in the chamber, and the motor was immediately activated. After a few seconds of operation, the motor was stopped and the mixture was visually inspected. The results indicated that the liquid became uniformly mixed within a short period of time. This confirms that the mixing mechanism is effective, which is critical since homogeneity is essential for the system's intended industrial function.

Rotational Motor Speed: How fast does the motor spin?

This test was designed to determine how fast the motor that drives the chamber mixing is rotating. It was conducted by first assembling the mechanism, attaching the motor to the base and the chamber, and then placing a tape "flag" on the top of the bottle. A video was then taken to record the number of times that the flag passed the same point, as a measure of RPM. In the end, we recorded 15 rotations in 5.24s and concluded that the motor spins the chamber at 170 RPM.

Success Criteria:

This prototype is designed to mix and transport liquid through a rotating chamber and pump system to demonstrate the first of two functioning parts of this project. Success will be measured by how well the device remains stable, mixes, and moves the target liquid from the bottle through the pump.

1. The first criterion is the stability of the prototype, which evaluates whether the bottle, motor, shell, and support pieces fit together and remain stable while the prototype is running. We can measure this by assembling the main parts and observing whether they remain supported with little wobble for at least 30 seconds when all devices are turned on.
2. The second criterion is the effectiveness of mixing, which looks at how well the spinning chamber mixes the liquid inside. We can measure this by adding food coloring to the liquid and turning on the motor. If the liquid becomes evenly mixed, visually uniform, within 30 seconds, the prototype meets the standard.
3. The last criterion is the transport performance of the liquid, which looks at whether the pump can successfully move the target liquid from the bottle to the nozzle. We can measure this by testing if 50 ml of liquid can be moved through the pump from the bottle in under a minute. On exhibit day, we can demonstrate this process simply by running the pump and demonstrating the liquid being moved.